

Subjective-Objective Emotion-Correlated Generation Network for Subjective Video Captioning

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Abstract—The emotional video captioning (EVC) task, which aims to generate factual descriptions based on the perceived subtle visual emotion cues, has received more and more attention and research. However, EVC is essentially an objective video captioning task, and ignores the subjective emotional reactions of video viewers, which cannot reflect personalized affective understandings of different viewers on the same video. To fill the research gap, we investigate the subjective video captioning (SVC) task in this paper, which aims to generate emotional captions by incorporating viewers' personalized emotional reactions upon the EVC task. SVC is extremely challenging, which lies in two aspects: 1) the correlative emotion perception between subjective and objective emotions and 2) the collaborative generation between emotional and factual information. To this end, we propose the Subjective-Objective Emotion-Correlated Generation Network (SO-ECGN) in this paper. Specifically, our SO-ECGN leverages the proposed dynamic mask attention and emotion domain shifting module to achieve the objective emotion incremental learning, and then, a subjective-objective emotions correlation module is proposed to adaptively combine two perspective emotions to provide accurate emotion guidance (*i.e.*, emotional polarity and intensity) for each generation step. Furthermore, an emotion-correlated decoder is proposed to generate subjective captions by adaptively referring to factual information and emotional information. Extensive experiments on three challenging datasets demonstrate the superiority of our approach and each proposed module, *i.e.*, reaching 79.2%, 45.1% on BLEU-1, CIDEr metrics on EmVidCap-L dataset.

Index Terms—Emotional video captioning, collaborative emotion perception.

I. INTRODUCTION

THE development of self-media platforms recently, such as TikTok, Twitter, and Facebook, has led to a surging

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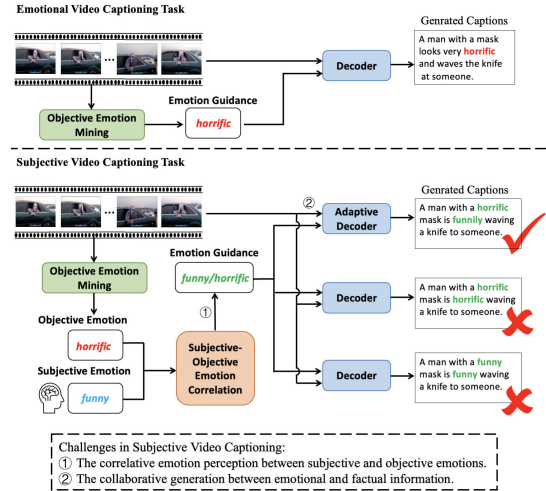


Fig. 1. Illustration of SVC task and our motivation, and comparison with the existing EVC task. SVC task takes the user's subjective emotions as inputs and collaborates them with the objective emotions implicit in the video to generate user-preferred subjective captions.

number of short videos created by users. There are a lot of emotions involved, such as the objective emotions contained in the video content and the subjective emotional reactions of the viewers. Thus, the field of intelligent video analysis pays more and more attention to the emotional understanding in the video [1], [2], [3], [4], [5]. Among intelligent video analysis [6], [7], [8], [9] tasks, the emotional video captioning (EVC) task [10], [11], [12], [13], which aims to describe factual content with the intrinsic emotions expressed in videos, has become an emerging topic.

However, the EVC task is essentially an objective video captioning task, while ignoring the crucial role of subjective emotional reactions by video viewers. To fill this gap, we argue that it is essential to study the Subjective Video Captioning (SVC) task, which is defined as 'Given a viewer-specified subjective emotion inputs, the SVC task aims to further incorporate viewer emotions based on the emotion and factual content implicit in the video to generate emotional descriptions'. Subjective emotion refers to the viewer's emotional response, while objective emotion refers to the consensus emotion by most viewers which is expressed by the main objects in the video. Consensus emotion is a collection of subjective emotions, overcoming the subjectivity of purely individuals and thus possessing a certain degree of objectivity. Thus, it is defined as objective emotion in this paper. We

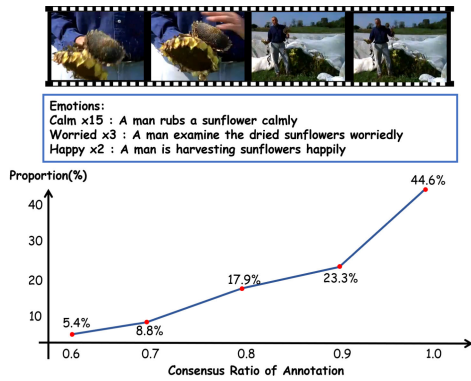


Fig. 2. Statistical analysis and example of consensus emotion distribution in EmVidCap-S dataset. The consensus ratio is calculated by normalizing the frequency of the emotion label per video.

conducted statistical analysis on the EmVidCap-S dataset, which is labeled by 20 viewers, and as shown in Fig. 2, 85.8% of the samples had a consensus rate higher than 0.8, which verifies the existence of consensus emotion. SVC has great application value for facilitating a more user-centric and emotion-aware video understanding and retrieval paradigm. For example, with the SVC techniques, the social media platform could provide personalized video descriptions [14], [15], [16], [17], [18], [19] and recommendations [20], [21], [22], [23], [24], [25] that better resonate with viewers' emotional preferences.

Subjective video captioning is extremely challenging, which lies in two aspects. Firstly, subjective and objective emotions are often inconsistent, which leads to the first challenge that the co-modeling of two perspective emotions is difficult. For example, as shown in Fig. 1, the man with a ghost mask is waving a knife horribly. However, when the man's behavior is considered to be a joke, the subjective emotion reaction of the viewer is more inclined to "funny". In doing so, the caption model should generate descriptions with "funny" as the dominant emotion, but should not ignore the objective visual emotional cue of "horrific" simultaneously. In this case, subjective and objective emotions are "funny" and "horrible", which are in conflict. Thus, how to correlative perceive and co-model subjective and objective emotions is the first key question to the SVC task. Secondly, vivid and diverse caption generation requires the coordination of factual semantics and emotional semantics. For example, when generating adjectives or adverbs, emotional semantics should have a greater guidance; whereas, when generating nouns or verbs, factual semantics should have more weights. Thus, how to coordinate factual semantics and emotional semantics adaptively in video caption generation is the second key question to the SVC task.

To fill the research gap, we propose the Subjective-Objective Emotion Correlated Generation Network (SO-ECGN) in this paper to adaptively generate user-preferred emotional captions word-by-wordly. Our SO-ECGN innovatively imitates the way humans generate opinions and descriptions of videos, namely, taking human subjective emotional reactions as the dominant and objective emotions expressed in videos

as the increment, and then, generating emotional expressions by an aggregation-and-disentanglement approach. Two main modules, *i.e.*, objective emotion incremental learning (OEIL) module and subjective-objective emotion correlation (SOEC) module, are proposed in our SO-ECGN in order to effectively address the two aforementioned challenges.

In our OEIL module, we aim to mine accurate and customized objective emotional cues expressed in the video content at each caption generation step. Firstly, we leverage a pre-defined emotion dictionary as the basis of emotion mining, which contains 179 emotion words based on a classical psychological theory *Plutchik's Wheel of Emotions* [26]. Subsequently, we propose a Dynamic Mask Attention (DMA) to perceive the emotional state of each generation step, which utilizes the alignment information between historical captions and emotion dictionary, since in word-by-word generation, the emotional needs, *i.e.*, emotional polarity and emotional intensity, of the current generation step can be determined from the previously generated text. However, there is a modality gap between the perceived emotions and the objective video emotional cues. Thus, an Emotion Domain Shifting is proposed before DMA to use video features to shift the whole emotion dictionary to align with the objective videos.

In our SOEC module, two perspective emotions are integrated and disentangled into the emotional polarity and intensity required at each generation step, which determines the emotional tendency to generate words, and how much emotional guidance is needed at the current generation step, respectively. To be specific, self-aggregation is utilized firstly to generate the self-weight to both two emotions, in order to facilitate the dynamic fusion. In disentanglement, two orthogonal pooling operations, *i.e.*, spatial-wise pooling and channel-wise pooling, are applied at first to the fused emotion to generate two decoupled factors. And then, the emotional polarity and intensity are disentangled by the element-wise multiplication between decoupled factors and fused emotions. Finally, the emotion-correlated generation leverages an adaptive decoder to balance emotional information and factual information in caption generation, generating vivid and diverse captions and leading to a promising performance. Extensive experiments on three challenging datasets demonstrate the superiority of our approach and each proposed module.

In short, our main contributions are summarized as follows:

- We propose a novel Subjective-Objective Emotion Correlated Generation Network (SO-ECGN) in this paper for the Subjective Video Captioning (SVC) task. Our SO-ECGN correlatively aggregates subjective and objective emotions and disentangles them to emotional polarity and intensity required by each generation step, providing the customized and accurate emotional guidance for each word generation.
- To achieve effective subjective-objective emotion correlation, we propose an objective emotion incremental learning module to firstly utilize historical subtitles and a domain-transferred emotion dictionary to generate dynamic masks, and then participate in the attention of video and emotion dictionaries to accurately mine objective video emotion cues.

- The emotion-correlated generation is proposed to balance emotional and factual information to generate vivid and diverse captions. Extensive experiments on three public benchmarks (*e.g.*, EmVidCap-S, EmVidCap-L, and EmVidCap) prove the effectiveness of our method and each proposed module, *e.g.*, reaching 79.2% and 45.1% on BLEU-1 and CIDEr, respectively, on EmVidCap-L.

II. RELATED WORK

A. Visual Emotional Analysis

With the popularity of a large number of videos on social platforms, researchers have attempted to understand the emotional cues contained in videos [27], [28], [29], [30], [31], [32], [33]. Existing visual emotional analysis can be divided into several categories. Firstly, most current work is dedicated to solving emotion classification task [34], [35], [36], [37], [38], [39], [40]. Researchers often focus on discriminative features (such as actions and expressions) in visual information to distinguish different emotions. Rao et al. [41] learn multi-level deep emotion representation through image semantics, image aesthetics and low-level visual features to predict emotion category. Zhang et al. [42] integrate hierarchical information of image semantics and aesthetics features to generate high-quality emotional representation. Furthermore, due to the diversity of human emotions, some researchers are devoted to visual emotion distribution learning instead of a single dominant emotion prediction [40], [43], [44], [45], [46]. Yang et al. [47] propose a circular-structured representation based on emotion circles to unified emotional state, which considers various attributes such as emotional polarity, intensity, etc. Zhang et al. [48] propose a temporal erasing network to locate keyframes and concentrate on the contexts that include complementary information via iteratively erasing keyframes.

Besides, many researchers attempt to leverage the powerful visual understanding capabilities of LVLMs to enhance visual emotional analysis [49], [50], [51], [52]. Cheng et al. [53] propose Emotion-LLaMA, a model that seamlessly integrates audio, visual, and textual inputs through emotion-specific encoders. By aligning features into a shared space and employing a modified LLaMA model with instruction tuning, Emotion-LLaMA significantly enhances both emotional recognition and reasoning capabilities. Zhang et al. [54] propose a novel attention mechanism, termed MODular Duplex Attention (MODA) to simultaneously conduct the inner-modal refinement and inter-modal interaction. Besides, a correct-after-align strategy is used to effectively decouple modality alignment from cross-layer token, significantly enhancing the capabilities in perception, cognition, and emotion tasks. Chen et al. [55] propose an EMotionally Omnipresent Voice Assistant (EMOVA), which enables LLMs to generate spoken dialogue with vivid emotions while maintaining the leading vision-language performance by a semantic acoustic disentangled speech tokenizer.

B. Traditional Video Captioning

Traditional video captioning task aims to generate objective and accurate factual sentences to describe videos. Nowadays,

most of the existing methods are sequential learning methods with encoder-decoder structure [56], [57], [58], [59], [60], [61], [62]. Researchers usually use attention-based encoders to capture semantic information in videos. With other carefully designed auxiliary information, video features are fed into auto-regressive decoders such as LSTM and Transformer to generate descriptions. Yang et al. [63] propose to utilize conceptual information to enhance video subtitles. Specifically, they proposed a concept-aware video subtitles framework to detect conceptual information in videos through video-to-text retrieval and multimedia features of videos, and designed a hybrid attention module to leverage concepts and video content to influence local decoding. Gao et al. [64] propose to utilize auxiliary tasks to enhance caption learning. On the one hand, the cross-modal matching task could learn hierarchical representations of videos under the guidance of three-level representations of language. On the other hand, the grammar guidance task helps to obtain grammar-aware video captions, thereby generating high-quality captions that are not only globally relevant to the video content but also similar and syntactically consistent with the real description. Jia et al. [65] propose to enhance video captioning by using only synthetic text data, which specifically leverages the exceptional text generation capabilities of LLMs to produce high-quality and diverse captions tailored to the target domain. Besides, they introduce Mixture of Scale and Shift experts (MoS²) to effectively utilize these captions, which employs routing networks to dynamically allocate tokens to appropriate experts, enabling tailored responses to input variations. Liang et al. [66] propose to leverage concise and easily-learned image samples to further diversify video samples, which compels the captioner to learn more effectively against redundancy and ambiguity. Besides, they design a Visual-Semantic Guided Captioning component to guide the network in flexibly learning ground truth captions based on the compounded samples. Rao et al. [67] propose a collaborative multi-modal graph network (CMGNet) to explore the interactions among multi-modal features, which designs a Basis Vector Compression module to reduce the redundant nodes in graphs and thus improve the efficiency, and an Intra-inter Attentive Graph to improve the graph representation by sharing information across intra-andinter nodes.

C. Emotional Video Captioning

Traditional video captioning tends to focus on the accuracy of factual description, thereby ignoring the learning of emotions in user-generated videos, which makes the generated sentence a bit boring and soulless. Therefore, some researchers propose emotional video captioning task, which mines emotional clues hidden in videos to generate more vivid captions. Wang et al. [11] firstly propose this task and release a new dataset. They design two independent prediction networks for factual and emotional captioning separately. The final output is generated by weighted average. Song et al. [68] propose a contextual attention network to recognize and describe the fact and emotion in the video by semantic-rich context learning. Song et al. [10] propose a novel tree-structured emotion learning module to achieve explicit emotion perception. Song et al. [13] incorporate visual context, textual context,

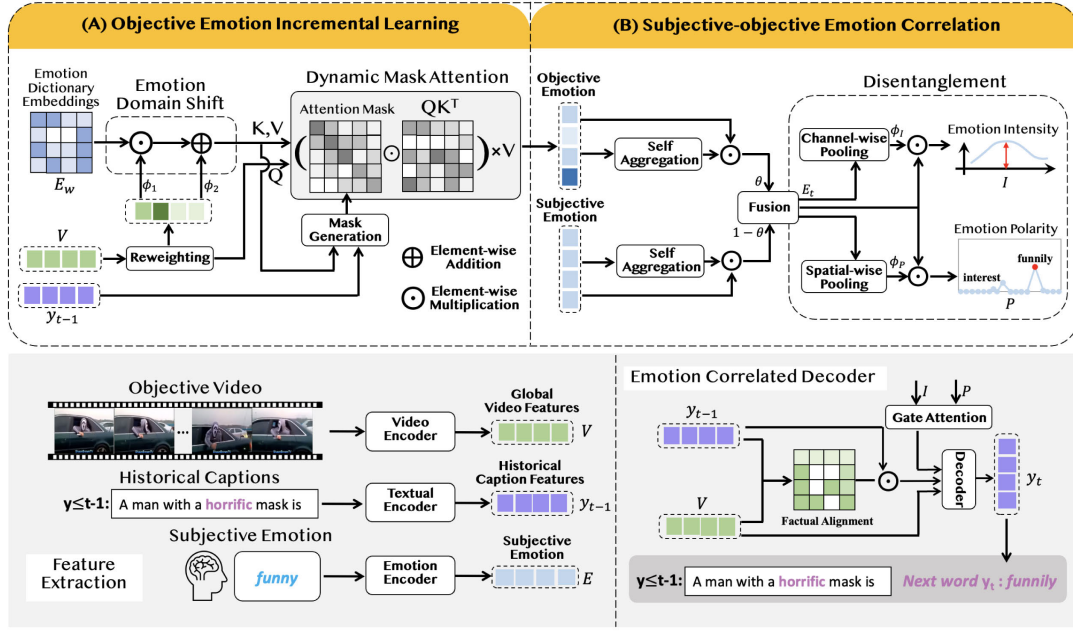


Fig. 3. The overview of our proposed subjective-objective emotion correlation network. It mainly consists of the objective emotion incremental learning, the subjective-objective emotion correlation, and the emotion-correlated decoder.

and visual-textual relevance into an aggregated multimodal contextual vector to enhance video captioning. Ye et al. [12] propose a dual-path collaborative generation network to address the challenge of dynamic emotion changes during generation. Moreover, an emotion-adaptive decoder is designed to solve the problem of overemphasis on the role of emotional guidance.

Although existing EVC methods pay attention to the importance of video emotional cues, they are still based on objective caption generation of video content and lack consideration of the subjective emotional reactions of viewers. Compared with EVC, our proposed SVC task has significant differences in terms of motivation, pipeline, and technical challenges. Firstly, in terms of motivation, EVC is motivated by the need for emotional expression in the description of traditional video captioning task. Unlike EVC, SVC takes into account the different emotional responses of different viewers to the same video. In addition to solving the problem of emotional needs in descriptions, SVC could also generate diverse emotional expressions based on users' subjective emotions, which is a crucial capability that EVC could not achieve. Secondly, in terms of pipeline, EVC only needs to extract emotional cues from visual content, while SVC requires additional input of the user's subjective emotions. This also leads to a third difference between the two tasks in technical challenges. The key challenge of EVC lies in accurately capturing visual emotional cues and coordinating factual-emotional semantics. Building upon this, SVC also needs to effectively integrate subjective and objective emotional cues, especially in scenarios involving conflict between them. Therefore, we believe that SVC has significant differences and faces greater challenges than EVC.

III. METHODOLOGY

Subjective Video Captioning (SVC) aims to describe user-preferred factual content with the intrinsic objective emotions expressed in videos and subjective emotional reactions by reviewers. The global video content and the subjective emotion are given in SVC task. To fill the research gap, we propose the Subjective-Objective Emotion Correlated Generation Network (SO-ECGN) in this paper, which mines objective emotions in videos at first, and then correlate with subjective emotions to generate emotional captions. The overall framework of our method is illustrated in Fig. 3, which mainly consists of three parts: 1) Objective Emotion Incremental Learning, 2) Subjective-objective Emotion Correlation, and 3) Emotion-correlated Generation.

A. Feature Extraction

1) *Video Encoder:* Following previous work [10], [11], we first downsample the video and leverage the pre-trained models to extract appearance features F_a and motion features F_m of the video, respectively. Then, we leverage two linear layers to further encode these two features. Finally, we use a standard self-attention to capture inter-frame dependencies to get the final compact video-level features $V \in \mathbb{R}^{N_T \times d_v}$.

$$F = [\phi_f^1(F_a); \phi_f^2(F_m)], \quad (1)$$

$$V = \text{Atten}_v(F)_{(Q,K,V):F} \in \mathbb{R}^{N_T \times d_v}, \quad (2)$$

where $[\cdot]$ is the concatenate operation, Attn is the attention operation and ϕ_f^1, ϕ_f^2 denotes two linear layers. The obtained video features V capture factual semantics, e.g., scenes, objects, and the emotions contained between visual elements.

2) *Subjective Emotion Encoder*: For each video, we support a multi-emotion input from users' subjective view. Specifically, we define a subjective emotion word set as $\Omega = \{s_i\}_{i=1}^{N_s}$, where s_i represents the i -th subjective emotional word, such as "happy" and "sad". Subsequently, we use a pre-trained word embedding model to encode each emotion word and take the average of all embeddings as the overall subjective emotion feature E :

$$E = \frac{1}{N_s} \sum_{i=1}^{N_s} s_i \in \mathbb{R}^{N_s \times d_e}, \quad (3)$$

where N_s is the number of subjective emotion words. Subjective emotion E is a key factor in subjective video caption generation, which it is also regarded as the basis and domination for subjective-objective emotion correlation.

B. Objective Emotion Incremental Learning

The focused objective visual emotions are constantly changing during word-by-word caption generation. Perceive the subtle changes of objective emotions in the video is the key to generate accurate emotional captions. To this end, we propose an objective emotion incremental learning module, which contains two sub-modules, *i.e.*, emotion domain shifting and dynamic mask attention, helping to perceive objective emotion increments at each time-step.

1) *Emotion Domain Shift*: Previous methods usually directly leverage a predefined emotion dictionary to obtain visual emotional cues in video contents. However, directly encoding the emotional feature with text embeddings of emotional words ignores the gap between text and visual emotion modality. To fill the gap, we propose to leverage visual features to perform emotion domain shift at first.

Meanwhile, we notice that not all video contents are helpful in mining emotional clues at current generation step. Different temporal segments of the video contain different objective emotions. Thus it is of great help to accurately locate the video moment corresponding to the current generation step for obtaining objective emotional increments. Besides, for different generation steps, the model has different emotion requirements due to different contexts. Therefore, it is necessary to focus on the relevant video frames and ignore irrelevant video frames in the current generation step, that is to perform **adaptive video reweighting**. Specifically, we propose to use historical caption features, which reflect the current generation state, to predict temporal weights and further perform adaptive reweighting. We firstly leverage a cross attention to align historical caption features Y^{t-1} with video features V . Then, we concatenate both two features, and project to obtain the temporal weights R^T . Finally, we leverage R^T to achieve reweighting:

$$\tilde{Y}^{t-1} = \text{Atten}_R(V, Y^{t-1})|_{Q:V, [K,V]:Y^{t-1}} \in \mathbb{R}^{N_T \times d_t}, \quad (4)$$

$$R^T = \text{ReLU}(\phi_r([V; \tilde{Y}^{t-1}])) \in \mathbb{R}^{1 \times N_T}, \quad (5)$$

$$\tilde{V}^t = R^T \odot V \in \mathbb{R}^{N_T \times d_v}, \quad (6)$$

where ReLU is the activation function and $\odot$ is the element-wise multiplication. By selectively reweighting the temporal information based on aligned historical caption features, the

model could focus on the most relevant frame and mine the most accurate objective emotion increment at each time-step.

Inspired by the approach in conventional transformer [69], [70], we design two dynamic modulation parameters by reweighted video features to perform emotion domain shift, which could effectively control the shifting direction. Specifically, we use MLP to predict these two parameters:

$$\alpha = \phi_{ds}^1(\tilde{V}^t), \quad \beta = \phi_{ds}^2(\tilde{V}^t), \quad (7)$$

where ϕ_{ds}^1 and ϕ_{ds}^2 denote MLP transformation. Subsequently, we normalize a pre-defined emotion dictionary [10] $E_w = \{e_i\}_{i=1}^{N_w}$, where e_i denotes the word embedding of the i -th natural emotion word, such as "glad". It manually builds an emotion tree with 179 emotion words based on a classical psychological theory *Plutchik's Wheel of Emotions*. We use dynamic modulation parameters to shift it to real visual emotion space:

$$\tilde{E}_w = \alpha \odot \text{Norm}(E_w) + \beta \in \mathbb{R}^{N_w \times d_e}, \quad (8)$$

where Norm denotes normalization operation.

2) *Dynamic Mask Attention*: Inspired by [10], we propose the Dynamic Mask Attention module to further mine the objective emotion increments at each generation step. After obtaining the visual-aligned emotion dictionary, and considering historical caption features could reflect the emotional requirements of the current time step, we leverage and project the historical caption features with a shifted emotion dictionary to obtain a dynamic mask. The formulation is:

$$M = \delta(\phi_m^1 \tilde{E}_w + \phi_m^2 Y^{t-1}), \quad (9)$$

where δ and ϕ denote the sigmoid activation function and linear transformation, respectively. It is worth noting that our emotional mask is soft and does not block out all impossible emotions. Subsequently, we mine objective emotion increments from the reweighted video features by mask attention mechanism. The formulations are:

$$\gamma_{i,j} = \frac{m_{i,j} \cdot \exp(\tilde{V}_i \tilde{E}_{w,j}^T / \sqrt{d})}{\sum_k m_{i,k} \cdot \exp(\tilde{V}_i \tilde{E}_{w,k}^T / \sqrt{d})}, \quad (10)$$

$$\Delta E^t = \phi_e^1(\text{Pooling}(\phi_e^2(\mathcal{R}\tilde{E}_w))) \in \mathbb{R}^{N_T \times d_e}. \quad (11)$$

where $m_{i,j}$, $\gamma_{i,j}$ are elements of $M, \mathcal{R}$, respectively. ϕ_e^1 , ϕ_e^2 are two linear layers. Pooling is the average pooling operation. The obtained objective emotion cues effectively combine the semantic information in emotion words and the visual emotion information hidden in the video, which alleviates the gap between different modalities and provides accurate emotional guidance during caption generation.

C. Subjective-Objective Emotion Correlation

1) *Subjective-Objective Emotion Aggregation*: As aforementioned, our SOEC module regards subjective emotions and objective emotions as the basis and increment, respectively, which is similar to human generate opinions and descriptions of videos that human subjective emotions should dominate. To this end, we first leverage self-aggregation to generate weights of subjective and objective emotions, and then apply self-weight to two emotions in order to better fusion. Afterward,

the aggregation is achieved by weighted combination of these two emotions. The formulation are:

$$G_S = \text{Pooling}(\phi_c^1(E)), \quad G_O = \text{Pooling}(\phi_c^2(\Delta E^t)), \quad (12)$$

$$E^t = \theta(G_S \odot E) + (1 - \theta)(G_O \odot \Delta E^t), \quad (13)$$

where Pooling and ϕ denote average pooling operation and linear transformation respectively, and θ is a hyper-parameter to balance two types of emotions. The combined emotion representation is mainly based on the user's subjective emotion E , ensuring that the generation is user-preferred. Meanwhile, it also learns adaptive objective emotion increments at each generation step, reducing the bias between the current generating state and the factual video content.

2) *Emotion Disentanglement*: Emotion guides the generation of video captions in various forms (*i.e.*, intensity and polarity). Emotional polarity controls the emotional color of generated words, and emotional intensity reflects how much the current generation step requires emotional guidance. Thus, we disentangle the aggregated emotion representations obtained at each generation step into emotion polarity P^t and emotion intensity I^t . Formally, we leverage average pooling operations in spatial and channel dimensions to obtain the intensity and polarity disentanglement factors:

$$\mathcal{F}_I = \delta \left(\phi_I \left(\frac{1}{N} \sum_i E^t(i, :) \right) \right), \quad (14)$$

$$\mathcal{F}_P = \delta \left(\phi_P \left(\frac{1}{d_e} \sum_j E^t(:, j) \right) \right), \quad (15)$$

where ϕ and δ denote linear layer and sigmoid activation function, respectively. Subsequently, we leverage factors to disentangle emotion features into components in different directions by element-wise multiplication:

$$I_t = \mathcal{F}_I \odot E^t, \quad P_t = \mathcal{F}_P \odot E^t, \quad (16)$$

Disentanglement not only improves the interpretability of emotion guidance for caption generation, but also filters out the noise in the overall emotion features, further ensuring the accuracy of emotion guidance.

D. Emotion-Correlated Generation

During the word-by-word caption generation, we obtain the visually correlated text feature $\mathcal{T}^t$ to enforce factual alignment. Technically, we firstly calculate factual alignment M^t between historical captions and video features. Then the alignment scores are normalized to obtain the visually correlated text feature $\mathcal{T}^t \in \mathbb{R}^{N \times d_t}$.

$$M^t = \text{Tanh}(W_m Y^{t-1} + H_m V + b_m), \quad (17)$$

$$\mathcal{T}_i^t = \sum_{j=1}^t \text{Softmax}(M^t) Y^{t-1}, i \in [1, N_T]. \quad (18)$$

At the decoding step t , we aggregate all the sufficient semantics and use emotion intensity estimation to adapt the features. Finally, we obtain the probability distribution $P(y_t) \in \mathbb{R}^D$ of the next word y^t , which is defined as:

$$X^t = [V, \mathcal{T}^t, I^t \odot P^t] \in \mathbb{R}^{N_T \times (d_v + d_t + d_e)}, \quad (19)$$

$$h^t = \text{Decoder}(\text{Pooling}(X^t), h^{t-1}), \quad (20)$$

$$P(y_t) = W_p h^t + b_p, \quad (21)$$

where Decoder is a standard attention-based LSTM to generate the caption sentence step-by-step, and W_p and b_p are learnable parameters, respectively.

To ensure the correctness and diversity of the generated captions, we leverage two objective functions to guide the correct generation direction. Firstly, we leverage classic cross-entropy loss $\mathcal{L}_{ce}$ to ensure the correctness of word-by-word caption generation:

$$\mathcal{L}_{ce} = - \sum_t \log P(y_t). \quad (22)$$

Secondly, we leverage a contrastive learning loss for emotion verification. Specifically, we take the input video as the positive sample and choose a video with a different emotion category in the training batch as the negative sample. Given video features V^t and its generated caption features Y^t at the t -th decoding step, we calculate the positive alignment score $r_{i,j}^t$ between positive video and generated caption features like Eq. (17). The negative alignment score $\tilde{r}_{i,j}^t$ could be calculated in the same way between negative video and caption features. Finally, the contrastive loss is calculated as follows, and our model can be trained end-to-end by minimizing the weighted summation of two losses:

$$\mathcal{L}_{ctr} = -\log \left[\delta \left(\sum_t \sum_{i,j} r_{i,j}^t \right) \right] - \log \left[1 - \delta \left(\sum_t \sum_{i,j} \tilde{r}_{i,j}^t \right) \right], \quad (23)$$

$$\mathcal{L} = \lambda_{ce} \mathcal{L}_{ce} + \lambda_{ctr} \mathcal{L}_{ctr}, \quad (24)$$

where δ is a sigmoid function, λ_{ce} and λ_{ctr} are two hyper-parameters that aim to control the balance of two losses.

IV. EXPERIMENTS

In this section, we first illustrated the Experimental Preparation, including Datasets and Metrics, Implementation Details, and Baselines. Secondly, we compared with the existing state-of-the-art methods on all three datasets. After that, we conducted ablation studies to discuss the effectiveness of different modules and sub-modules, as well as some important hyper-parameter settings. We further analyzed the Qualitative Results and Emotional Accuracy. Finally, we compared the proposed method with the Large Vision-Language Models (LVLMs).

A. Experimental Preparation

1) *Existing Datasets*: We validate the performance of our model on three public benchmarks, *e.g.*, **EmVidCap-S** [71], **EmVidCap-L** [72], and **EmVidCap** [11]. EmVidCap-S contains 240/134 videos and 8,169/4,611 sentences for training/testing from the traditional captioning dataset MSVD [71]. EmVidCap-L is built based on a traditional emotional video prediction dataset VideoEmotion-8 [72]. It is divided into 1,141/382 videos and 19,398/6,527 sentences for training/testing, respectively. EmVidCap is the combination of EmVidCap-S and EmVidCap-L, which contains 1,381/516 videos and 27,567/11,138 sentences for training/testing.

2) *New Dataset Construction*: Additionally, we construct a new dataset named EmVidCap-H. We collect a series of hard example videos from EmVidCap [11]. The video could be regarded as a hard example, in which objective emotions reflected from video contents conflict with subjective emotions. The remaining samples of the EmVidCap dataset are named as **EmVidCap-R**. Specifically, we organize 10 annotators to evaluate whether the objective emotions in the video conflict with the emotions in the ground-truth sentences and then make a judgment based on the results agreed by the majority. Finally, we collect 588/263 samples for training/testing respectively, which reflects a strong subjective color and helps to highlight the advantages of our model. It is worth noting that for the important subjective emotion component, due to the emotion conflict of these hard samples, we directly use the emotional words in the ground truth sentences as subjective emotional inputs instead of additional annotations. Besides, during the training phase, we select the consistent subjective emotions that are shared by most annotators for each video to ensure the optimizability and convergence of our model and avoid disagreements among annotators, which is a common strategy to disambiguate subjective emotion labels [10], [11], [12], [13]. However, during the testing phase, we could input different subjective emotions to evaluate the diversity of the generated captions.

3) *Metrics*: For the subjective video captioning task, four common reference semantic evaluation metrics are used to evaluate the quality of generated captions. **BLEU-4** calculates the similarity between generated captions and ground-truth based on n-gram overlap. **METEOR** computes exact matches and word order matches score. **ROUGE-L** counts overlapping words, phrases and sequences. **CIDEr** to assign different weights to different n-grams based on TF-IDF, which is closer to the human standard for judging sentence similarity. In addition, a caption-to-visual content evaluation metric **CLIP-Scores** is used to measure the semantic similarity between the generated caption and the video. For emotional video captioning task, we additionally add two accuracy metrics Acc_{sw} , Acc_c [11] to evaluate emotion accuracy on word-level and sentence-level.

More importantly, two hybrid metrics BFS and CFS that combine BLEU and CIDEr with emotion metrics are used since both factual and emotional semantics benefit the descriptions. They are calculated specifically as follows:

$$BFS = k \cdot \sum_{n=1}^4 \omega_n \cdot BLEU_n + (1 - k) \left(\frac{Acc_{sw} + Acc_c}{2} \right), \quad (25)$$

$$CFS = k \cdot CIDEr + (1 - k) \left(\frac{Acc_{sw} + Acc_c}{2} \right), \quad (26)$$

k is a harmonic factor which is set to be $k = 0.8$ in practice. The parameter ω_n is the assigned weight for BLEU- n ($n = 1, 2, 3, 4$), which is set as $(\omega_1, \omega_2, \omega_3, \omega_4) = (0.1, 0.2, 0.3, 0.4)$ since BLEU- n is more valuable as n increases.

4) *Implementation Details*: For a fair comparison, we downsample each video to $N_T = 30$ keyframes as video inputs and extract appearance features by leveraging ResNet-101 or ResNet-152 [74] and motion features using

3D-ResNext-101 [75]. Besides, we test the model with the modern CLIP [76] features. For captions, we limit the maximum length of caption to 15. The sizes of overall vocabulary that contains all words in the corpus for three datasets are 9,637, 13,980, and 14,034, respectively. For emotion dictionary, there are $N_w = 179$ emotion words. All emotion word embeddings are encoded by GloVe [77]. The feature dimension is set to $d = d_v = d_t = d_e = 300$. We set $\theta = 0.7$ to balance two types of emotions. The hidden size of the caption decoder is set to $h = 512$. For training, we adopt the Adam optimizer with a learning rate of $7e-4$, the batch size is set to 128, and λ_{ce} and λ_{ctr} are set to 1 and 0.2, respectively.

5) *Baselines*: In this work, we consider several state-of-the-art methods as baselines used against our SO-ECGN.

- SA-LSTM [73] (CVPR 2018) proposes a reconstruction network to use both the forward (video to sentence) and backward (sentence to video) flows for video captioning.
- FT [11] (TMM 2022) proposes a dual-stream generation network, which design two independent prediction stream for factual and emotional captioning separately.
- CANet [68] (TMM 2023) recognizes and describes the fact and emotion by semantic-rich context learning. In addition, they design a contrastive loss to encourage discriminative feature learning from video-caption pairs.
- EPAN [10] (ACM-MM 2023) proposes a tree-structured emotion learning module to learn visual emotion cues through psychology-level prior knowledge. In addition, a subordinate emotion masking mechanism is developed to achieve coarse-to-fine emotion learning.
- VEIN [13] (TIP 2024) proposes a vision-based emotion interpretation network, which incorporates visual context, textual context, and visual-textual relevance into an aggregated multimodal vector to enhance video captioning.
- DCGN [12] (ACM-MM 2024) proposes a dual-path collaborative generation network, which dynamically perceives visual emotional cues evolutions and leverages emotion intensity estimation to determine the involvement of emotional guidance.

Most compared baselines, *i.e.*, VEIN, EPAN and DCGN, are from Emotional Video Captioning task. In addition, we also compare some advanced video captioning methods, *i.e.*, FT, CANet, and SA-LSTM, to further illustrate our superiority. *It is worth noting, to make a fair comparison, we input the subjective emotions for all these baselines as the same as our model, and reconstruct all results.*

B. Main Comparison

Table II shows the comparison between our model and state-of-the-art methods. The proposed SO-ECGN (Ours) achieves the best performances on all datasets under all three feature extractions, *i.e.*, ResNet-152, ResNet-101 + 3D-ResNext-101, and CLIP.

- Compared with traditional video captioning methods, our method achieves significant improvements on all metrics. For instance, on EmVidCap-S dataset, our method brings remarkable improvements, *i.e.*, +12.1%, +18.2%, +9.4%

TABLE I

Name	MORE DETAILS ABOUT EACH DATASET # of Videos	Duration (s)	Annotations
EmVidCap-S	240/134	9	8169/4611
EmVidCap-L	1141/382	107	19398/6527
EmVidCap	1381/516	87	27567/11138
EmVidCap-H	588/263	61	11737/5677

improvements on BLEU-3, METEOR, ROUGE-L over SA-LSTM with CLIP features. The reason for such huge improvements is that compared with traditional video captioning methods, our method dynamically perceives visual emotional cues and disentangles emotional intensity and polarity to guide caption generation, making the captions more interesting and vivid.

- Our model also achieves better performance than the state-of-the-art EVC methods. For example, on EmVidCap-L dataset, the improvements are + 9.0%, + 5.6%, + 18.1% than DCGN on METEOR, ROUGE-L, and CIDEr with ResNet101 + 3D-ResNeXt-101 features. The reason is that our method correlates subjective and objective emotions on the basis of mining objective visual emotions well. In contrast, the EVC methods cannot handle the situation that subjective and objective emotions are inconsistent and provide accurate emotion guidance for each generation step.
- On EmVidCap dataset, our method also outperforms the state-of-the-art methods on almost all metrics. One exception is that DCGN performs slightly better on METEOR with ResNet101 + 3D-ResNeXt-101 features. One possible reason is that our emotion domain shifting module may lead to diverse emotional shifting space that are not limited to ground truth.
- On EmVidCap-Hard dataset, our model demonstrates more significant advantages. Videos in EmVidCap-Hard have emotional conflicts between video contents and textual annotations. Previous methods are difficult to collaborate inconsistent emotional cues thereby generating poor captions. Instead, our SOEC module effectively collaborates subjective and objective emotions through adaptive emotion aggregation and disentanglement and captures more accurate emotion guidance. As shown in Table III, compared with results on original EmVidCap dataset, previous methods suffer significant performance reduction of at least 5% for all metrics. In contrast, our model only shows a slight performance reduction, *i.e.*, 2.0% and 2.2% reductions on BLEU-1 and BLEU-2 with ResNet-152 features, which further demonstrates that our model could generate more user-preferred captions.
- Furthermore, except to above NLG-based metrics, we also leverage CLIP Scores to evaluate the semantic similarity between the captions generated by our model and the video content. As illustrated in Table IV, we observe that our model surpasses all existing methods on CLIP score, *i.e.*, 3.6%, 5.2%, 4.1% improvements than DCGN on three datasets, respectively, which demonstrates that our method could fully understand the video content, extract its semantic information and convert it cross-modally into captions.

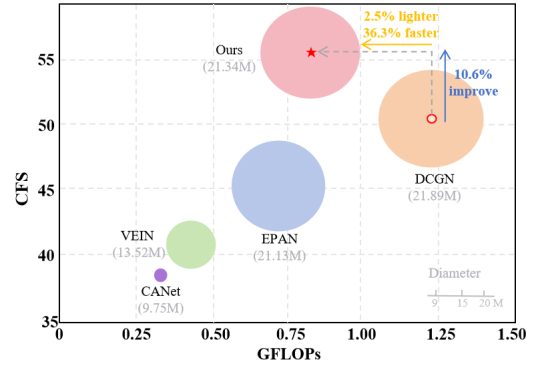


Fig. 4. Comparison with state-of-the-art methods on computational complexity (GFLOPs), trainable parameters of model, and performance on CFS metric on the EmVidCap-H dataset.

1) Model Parameter and Computational Complexity Evaluation: The efficiency and scalability are an important factor to measure the superiority of a model. Thus, we add an experiment to compare our model with state-of-the-art methods on computational complexity, number of parameters, and performance. The results are shown in Figure 4. Firstly, we can observe that compared to the baseline model EPAN, we achieve a 24.9% performance improvement with only an additional 0.21M parameters. Besides, compared to the most powerful method DCGN, our model brings 10.6% improvement on CFS with 2.5% lighter parameters, and 36.3% faster computation. These comparisons show that our model takes both performance and computational costs into account in a comprehensive way. Our approach brings performance improvements, while the additional computing consumption is relatively low. Besides, we also make a comparison with state-of-the-art methods on the memory sizes and the average execution times to generate a caption. As shown in Table V, our method shows competitive performance on these two metrics, which proves the potentiality and value of our model.

C. Ablation Studies

1) Discussion on Each Proposed Module: To further investigate the contribution of each component in our method, we conduct a set of ablation studies. The results are shown in Table VI. Components OEIL and SOEC represent objective emotion increment learning and subjective-objective emotion correlation, respectively. Firstly, we observe that each component could independently improve the performance of our model. For example, our OEIL module brings *i.e.*, + 9.0%, + 16.9% improvements on METEOR and CIDEr, respectively. This is because our OEIL module first performs emotional domain shifting to eliminate the gap between textual and visual emotion modalities, and then retrieves the video segments corresponding to the current generation step and leverages them as emotion masks to mine emotional cues from the open psychology emotion dictionary. Our SOEC module brings + 4.7%, + 6.0% improvements on BLEU-3 and ROUGE-L, respectively. This is because our SOEC module correlates subjective and objective emotion by adaptive reweighting and disentangling the correlated emotion into

TABLE II
COMPARISON WITH THE STATE-OF-THE-ART METHODS IN SUBJECTIVE VIDEO CAPTIONING TASK ON THREE CHALLENGING DATASETS, I.E., EMVIDCAP-S, EMVIDCAP-L, AND EMVIDCAP. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Datasets	Methods	Backbones	BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE-L↑	CIDEr↑
EmVidCap-S	FT [11]	R152	77.2	60.3	47.4	36.3	29.0	63.4	62.5
	CANet [68]		78.0	62.1	50.0	38.6	29.8	63.5	63.3
	VEIN [13]		81.5	66.4	55.3	45.2	32.7	67.1	71.6
	EPAN [10]		81.7	66.8	55.6	45.8	33.4	67.5	72.4
	DCGN [12]		83.2	69.1	57.8	47.2	34.4	71.1	73.9
	Ours		86.1	71.4	58.9	48.0	36.9	74.1	82.8
	CANet [68]	R101+RN	78.5	64.0	52.1	41.8	30.8	65.7	74.4
	VEIN [13]		81.4	66.6	56.0	44.8	33.9	67.4	76.6
	EPAN [10]		81.5	67.2	56.3	45.1	34.8	67.8	77.9
	DCGN [12]		83.8	69.4	58.1	47.6	34.9	71.4	79.4
	Ours		85.9	71.7	59.3	48.3	36.4	73.6	82.6
	SA-LSTM [73]	CLIP	80.7	67.9	56.3	45.5	33.0	68.2	72.1
	VEIN [13]		84.1	71.3	60.3	48.9	34.9	70.7	82.7
	EPAN [10]		84.8	71.8	61.0	49.4	36.2	71.3	83.5
	DCGN [12]		87.0	73.2	61.7	50.9	37.1	72.6	87.7
	Ours		89.5	74.9	63.1	52.0	39.0	74.6	89.6
EmVidCap-L	CANet [68]	R152	66.0	44.6	29.1	18.8	17.7	37.7	23.9
	VEIN [13]		68.7	47.5	32.1	21.1	18.8	40.4	27.4
	EPAN [10]		69.6	48.9	33.4	21.3	19.1	40.5	28.9
	DCGN [12]		71.5	49.5	35.6	22.8	20.4	41.8	30.1
	Ours		73.9	52.5	36.2	23.7	21.6	43.7	33.9
	CANet [68]	R101+RN	66.9	44.8	29.3	19.3	18.2	37.9	23.3
	VEIN [13]		68.6	47.7	31.9	21.3	18.8	39.2	26.8
	EPAN [10]		69.2	47.6	32.7	21.9	18.9	39.4	28.4
	DCGN [12]		71.1	48.9	35.2	22.7	20.1	43.1	29.3
	Ours		73.5	52.3	36.4	24.1	21.9	45.5	34.6
	SA-LSTM [73]	CLIP	71.0	51.1	34.5	22.5	19.6	40.7	30.2
	VEIN [13]		74.4	54.6	38.1	27.5	21.7	44.8	35.9
	EPAN [10]		74.9	55.1	38.7	27.8	21.9	46.0	37.4
	DCGN [12]		75.8	56.2	40.6	28.7	24.0	48.6	43.1
	Ours		79.2	58.9	41.6	29.5	24.3	51.5	45.1
EmVidCap	FT [11]	R152	67.6	47.2	32.0	21.6	20.4	43.1	29.0
	VEIN [13]		67.9	48.2	32.7	22.4	20.3	43.8	32.8
	EPAN [10]		69.2	49.5	33.4	22.7	21.0	44.1	33.5
	DCGN [12]		71.2	51.4	35.8	24.1	23.2	45.8	35.2
	Ours		74.2	54.9	40.1	27.2	23.2	48.9	38.7
	CANet [68]	R101+RN	68.1	47.7	32.9	22.5	19.7	43.7	34.5
	VEIN [13]		71.7	51.2	36.9	25.8	21.4	46.8	37.9
	EPAN [10]		72.1	51.9	37.5	26.1	21.8	47.2	38.4
	DCGN [12]		73.7	54.3	39.7	27.4	24.2	49.9	42.7
	Ours		74.5	55.2	40.7	27.9	24.1	50.6	44.8
	SA-LSTM [73]	CLIP	70.6	51.4	36.7	25.4	21.0	45.9	38.8
	VEIN [13]		75.0	54.9	38.9	27.4	22.6	48.3	42.2
	EPAN [10]		76.7	56.4	40.6	28.8	23.7	48.5	44.2
	DCGN [12]		77.2	58.7	41.7	29.2	26.1	54.2	51.3
	Ours		79.8	59.8	42.4	29.5	26.6	58.0	52.9

TABLE III
COMPARISON WITH THE STATE-OF-THE-ART METHODS IN SUBJECTIVE VIDEO CAPTIONING TASK ON EMVIDCAP-HARD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Backbones	BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE-L↑	CIDEr↑
FT [11]	R152	63.1	44.0	29.7	19.9	19.6	41.7	27.4
VEIN [13]		63.9	44.5	30.1	20.3	20.0	42.1	28.2
EPAN [10]		65.2	46.5	31.6	21.1	20.2	42.9	29.6
DCGN [12]		67.8	48.1	33.2	22.6	21.8	43.6	32.4
Ours		72.7	53.7	38.8	26.4	22.6	46.7	36.9
CANet [68]	R101+RN	63.6	44.1	29.5	19.6	19.8	42.2	31.7
VEIN [13]		65.1	45.8	31.2	21.0	20.5	43.1	33.6
EPAN [10]		66.4	47.4	32.3	21.6	20.9	43.8	35.9
DCGN [12]		68.2	48.5	33.4	22.5	22.0	46.0	39.8
Ours		73.0	53.9	39.1	26.8	23.2	48.8	42.7
SA-LSTM [73]	CLIP	67.4	47.3	32.5	22.2	20.2	44.6	35.1
VEIN [13]		70.4	49.9	35.0	24.5	21.1	46.9	38.8
EPAN [10]		71.5	51.0	35.9	25.0	21.6	47.3	40.1
DCGN [12]		72.8	52.0	36.6	25.5	22.4	51.2	45.7
Ours		77.8	57.4	41.3	28.8	25.4	56.8	51.6

emotion polarity and intensity. Both two are then to adaptively determine the extent to which emotional guidance participates in caption generation, thus bringing better performance. Finally, the combination of these two modules brings

TABLE IV

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON CLIP SCORE METRIC. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. $\Delta\%$ IS THE IMPROVEMENT

Datasets	Methods	CLIP Scores $\uparrow$	$\Delta\%\uparrow$
EmVidCap-S	VEIN [13]	60.9	-
	EPAN [10]	62.1	2.0
	DCGN [12]	68.8	13.0
	Ours	71.3	17.1
EmVidCap-L	VEIN [13]	52.4	-
	EPAN [10]	55.7	6.3
	DCGN [12]	59.1	12.8
	Ours	62.2	18.7
EmVidCap	VEIN [13]	56.1	-
	EPAN [10]	59.6	6.2
	DCGN [12]	63.3	12.8
	Ours	65.9	17.5

TABLE V

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE MEMOYSIZES AND THE AVERAGE EXECUTION TIME TO GENERATE A CAPTION

Method	Execution Times	Memory
CANet[4]	117.4ms	3346MB
VEIN[5]	152.9ms	4051MB
EPAN[1]	198.6ms	5602MB
DCGN[2]	247.8ms	6201MB
Ours	215.6ms	5887MB

+ 3.7% improvement on CIDEr and + 4.7% improvement on ROUGE-L than solely leveraging OEIL module and SOEC module, respectively, which further demonstrates the effectiveness of these two modules.

2) *Discussion on Each Sub-Modules in OEIL*: The OEIL module, which dynamically perceives the objective emotional increment to modify the emotional guidance of each time step, is one of the most important parts of our model. Therefore, we conduct ablation studies on the dense components of it. As shown in Table VII, our experiments are conducted under two different settings, i.e., w/ and w/o adaptive video reweighting (AVR). EDS and DMA denote emotion domain shifting and dynamic mask attention, respectively.

Firstly, we observe that without AVR, both EDS and DMA experience slight performance improvements for our model, i.e., 0.5%, 1.2% and 1.2%, 1.9% on BLEU-1 and CIDEr metrics for EDS and DMA, respectively. When adding AVR module, our model performs a remarkable performance improvement. We analyze that directly using global video features to perform shifting and capture emotion masks may lead to the wrong shifting direction and introduce noise contained in irrelevant frames to the current time step. In contrast, the AVR module retrieves the most relevant video segments to mine accurate objective emotion increments at the current time step, which brings remarkable improvements, i.e., 3.4% and 4.2% on METEOR and CIDEr metrics, respectively.

In addition, we discuss the impact of component orders. As shown in Table VIII, we observe that applying emotion shifting at first performs better for our model. Performing shifting on the global emotion dictionary effectively eliminates the gap between textual and visual emotion domains and helps to generate more accurate emotion masks, thereby bringing better

performance, i.e., 1.1% and 1.2% improvements on BLEU-1 and ROUGE-L metrics, respectively.

3) *Discussion on Dynamic Mask Attention*: In the process of generating attention masks, Y_{t-1} and $\tilde{E}_w$ are two important factors, which represents the historical caption feature and the visual emotion information, respectively. Thus, it is necessary to perform an ablation study to explore their roles. As shown in Table IX, removing $\tilde{E}_w$ and Y_{t-1} reduces the semantic and emotion accuracy metrics to varying degrees. Specifically, removing Y_{t-1} has a significant impact on semantic metrics. It determines the current generation state and provides powerful contextual information to ensure the coherence and logical flow of factual content and avoids unintended disruptions in the causal order of tokens. Without it, the model's ability to predict the next word decreases, which leads to a decrease in the accuracy and readability of generated captions. Meanwhile, removing $\tilde{E}_w$ has a significant impact on the emotion accuracy metrics. It captures the visual emotion information aligned with current video clips and provides a stable emotional anchor for the generation process and ensures the emotional controllability of the description. Without it, the model has difficulty bridging the gap between visual and textual emotions, resulting in inaccurate objective emotion mining.

Furthermore, we explore a series of more complex mask generation methods. Inspired by [81], we select four different attention mechanisms, namely self attention(SA), Transpose SA, Separable SA, and Additive Attention, respectively. Subsequently, we leverage the concatenation of $\tilde{E}_w$ and Y_{t-1} as the input of these attention mechanisms. As shown in Table X, they perform poorly in both factual and emotion accuracy metrics. We analyze that leveraging the concatenation of $\tilde{E}_w$ and Y_{t-1} directly will amplify the negative impact of the gap between text and visual information, especially when using complex attention mechanisms. Instead, we leverage two independent linear transformations to avoid confusion between two different modalities, thereby achieving accurate mask generation.

Overall, this construction allows our model to adaptively balance the influence of the visual emotion basis and the historical generation state on the current attention decision at each time step.

4) *Discussion on Emotion Domain Shifting*: In the emotion domain shifting module, we design two crucial parameters to perform shifting. It is necessary to perform an ablation study to explore their roles, respectively. Specifically, we remove shift and scale parameter respectively to observe the impact on the generated captions. As shown in Table XI, without the emotion domain shifting module, our model fails to capture the correct visual emotion cues to guide generation, which reduces the performance of our model. To be specific, we observe that using the scale parameter independently could only improve the model performance slightly, which only scales the features of the text emotion dictionary without performing any essential transformation. Instead, the shift parameter is the key element to bridge the gap between textual and visual emotion information, which brings a obvious improvement for our model.

5) *Discussion on Hyper-Parameter θ* : In subjective-objective emotion aggregation, we leverage a hyper-parameter

TABLE VI

THE RESULTS OF ABLATION STUDIES ON THE EMVIDCAP-L DATASET TO DISCUSS THE EFFECTIVENESS OF OUR PROPOSED COMPONENTS. OEIL AND SOEC REPRESENT THE OBJECTIVE EMOTION INCREMENTAL LEARNING MODULE AND SUBJECTIVE-OBJECTIVE EMOTION CORRELATION MODULE, RESPECTIVELY

Modules		BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE-L↑	CIDEr↑
OEIL	SOEC							
×	×	74.7	55.0	38.6	28.0	22.2	46.4	37.2
✓	×	77.4	57.6	40.8	28.7	24.2	50.2	43.5
×	✓	76.2	57.0	40.4	27.8	23.5	49.2	41.8
✓	✓	79.2	58.9	41.6	29.5	24.3	51.5	45.1

TABLE VII

THE RESULTS OF ABLATION STUDIES ON THE EMVIDCAP-L DATASET TO DISCUSS THE EFFECTIVENESS OF DENSE COMPONENTS IN THE OBJECTIVE EMOTION INCREMENTAL LEARNING MODULE. EDS AND DMA STAND FOR EMOTION DOMAIN SHIFT AND DYNAMIC MASK ATTENTION, RESPECTIVELY. THE ILLUSTRATION IS CONDUCTED UNDER TWO SETTINGS, I.E., W/ AVR AND W/O AVR. IT IS WORTH NOTING THAT AVR STANDS FOR ADAPTIVE VIDEO REWEIGHTING

	Components		BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE-L↑	CIDEr↑
	EDS	DMA							
w/o AVR	×	×	76.2	57.0	40.4	27.8	23.5	49.2	41.8
	✓	×	76.6	57.3	40.2	27.8	23.1	49.5	42.3
	×	✓	76.9	57.5	40.2	27.7	23.3	49.8	42.6
	✓	✓	77.6	57.9	40.8	28.2	23.5	50.4	43.3
w/ AVR	×	×	77.0	57.2	40.4	28.1	23.6	49.3	42.0
	✓	×	77.3	57.5	40.7	28.0	24.0	49.8	42.5
	×	✓	78.1	58.0	41.0	29.1	24.3	50.2	43.9
	✓	✓	79.2	58.9	41.6	29.5	24.3	51.5	45.1

TABLE VIII

THE EXPERIMENT RESULTS OF DIFFERENT ORDERS OF TWO COMPONENTS IN OEIL MODULE. DMA AND EDS REPRESENT DYNAMIC MASK ATTENTION AND EMOTION DOMAIN SHIFTING, RESPECTIVELY

Order	BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE-L↑	CIDEr↑
DMA→EDS	78.3	58.4	41.3	29.2	24.7	50.9	44.8
EDS→DMA (Ours)	79.2	58.9	41.6	29.5	24.3	51.5	45.1

TABLE IX

THE DETAILED ABLATION STUDY FOR DYNAMIC MASK ATTENTION ON EMVIDCAP DATASET

E_w	Y_{t-1}	BLEU-1↑	BLEU-4↑	EmVidCap METEOR↑	ROUGE-L↑	CIDEr↑	Acc_{sw} ↑	EmVidCap-H Acc_c ↑	BFS↑	CFS↑
×	×	77.3	28.0	24.0	49.8	42.5	72.6	71.2	47.7	53.1
✓	×	77.8	28.2	23.9	50.4	43.0	74.7	72.1	48.5	54.9
×	✓	78.5	29.1	24.3	50.8	44.1	73.2	71.8	48.8	55.4
✓	✓	79.2	29.5	24.3	51.5	45.1	75.6	73.4	49.4	56.2

TABLE X

THE COMPARISON BETWEEN OUR PROPOSED ATTENTION MASKS AND OTHER METHODS, WHERE SA DENOTES SELF ATTENTION MECHANISM

Method	BLEU-1↑	BLEU-4↑	EmVidCap METEOR↑	ROUGE-L↑	CIDEr↑	Acc_{sw} ↑	EmVidCap-H Acc_c ↑	BFS↑	CFS↑
SA [78]	77.1	27.4	23.5	49.5	42.3	71.4	70.6	45.8	52.0
Transpose SA [79]	77.6	28.2	23.8	50.2	42.9	73.8	72.2	47.0	53.9
Separable SA [80]	78.1	28.5	24.0	50.6	43.7	74.2	72.6	47.6	54.5
Additive Attention [81]	78.5	29.0	24.2	51.0	44.0	74.5	72.9	49.0	55.7
Ours	79.2	29.5	24.3	51.5	45.1	75.6	73.4	49.4	56.2

θ to balance two types of emotions, which has a significant impact for the quality of generated captions. Therefore, we conduct an experiment for θ in different settings. As shown in Table XII, the model achieves the best performance when $\theta = 0.7$, which matches our settings in the paper. We analyze that a too low value for θ results in a low proportion of subjective emotion and the generated captions

lack subjective color. On the contrary, too high value for θ may lead to an excessive proportion of subjective emotions, ignoring the objective emotional factors implicit in the video, which may cause the generated captions to lose some real emotional details reflected in the video. Therefore, the experiments prove the rationality of our settings for hyper-parameters.

TABLE XI
THE DETAILED ABLATION STUDY FOR EMOTION DOMAIN SHIFTING MECHANISM

Scale	Shift	BLEU-1↑	BLEU-4↑	EmVidCap METEOR↑	ROUGE-L↑	CIDEr↑	Acc _{sw} ↑	EmVidCap-H Acc _c ↑	BFS↑	CFS↑
×	×	78.0	28.7	23.7	50.5	43.2	74.0	72.3	47.2	54.3
✓	×	78.1	29.1	23.8	50.2	43.9	74.3	72.5	47.4	54.4
×	✓	78.5	29.3	24.0	50.9	44.1	74.8	73.0	49.1	55.9
✓	✓	79.2	29.5	24.3	51.5	45.1	75.6	73.4	49.4	56.2

TABLE XII
THE RESULTS OF THE EXPERIMENT FOR DIFFERENT HYPER-PARAMETER θ

θ	BLEU-1↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE-L↑	CIDEr↑
0.3	77.0	57.9	39.8	28.6	23.8	49.8	43.6
0.5	77.5	58.1	40.3	28.9	24.1	50.1	44.2
0.7	79.2	58.9	41.6	29.5	24.3	51.5	45.1
1.0	76.1	56.8	39.4	28.1	23.2	49.3	43.5

TABLE XIII
THE EXPERIMENTAL RESULTS ON EMOTION ACCURACY ON BOTH EMVIDCAP-H AND EMVIDCAP-R DATASETS

Dataset	Methods	Acc _{sw} ↑	Acc _c ↑	BFS↑	CFS↑
EmVidCap-H	VEIN [13]	54.4	52.6	40.6	41.7
	EPAN [10]	65.7	63.2	43.4	45.0
	DCGN [12]	72.1	70.5	45.4	50.8
	Ours	75.6	73.4	49.4	56.2
EmVidCap-R	VEIN [13]	63.4	62.4	42.4	43.6
	EPAN [10]	72.8	71.0	44.9	46.5
	DCGN [12]	77.4	75.6	46.4	51.9
	Ours	78.0	76.9	50.5	57.3

D. Subjective/Objective Emotion Accuracy

We leverage EmVidCap-H and the remaining of EmVidCap except EmVidCap-H as subjective and objective videos to measure the emotion accuracy. As shown in Table XIII, our method achieves accurate emotion prediction on both subjective and objective videos, which demonstrates our excellent performances in subjective and objective emotion learning, *i.e.*, + 15.1%/+ 16.1% improvements over EPAN on subjective videos with Acc_{sw} and Acc_c and + 8.8%/+ 10.4% improvements over DCGN on objective videos with BFS and CFS. Our model utilizes reweighted video features as mask to mine emotion cues from open psychology emotion dictionary and shifts it to real visual emotion space to achieve objective emotion increment learning at each time step, which could fully capture the subtle changes of objective emotions implicit the video. Meanwhile, the emotion disentanglement entirely disentangle emotions to guide caption generation clearly, which encourages to generate more accurate emotion words.

E. Human Evaluation

Since SVC is a highly subjective task, we add human evaluation to fully measure the quality of emotional descriptions. Inspired by the work [11], we designs four metrics, including (1) emotion accuracy: evaluate the accuracy of emotion in the sentence, (2) relevance: assess whether the sentence is relevant to the video content, (3) coherence: judge the logic and

TABLE XIV
THE RESULTS OF HUMAN EVALUATIONS

Metric	Accuracy	Relevance	Coherence	Usability
VEIN[5]	5.47	5.18	6.27	5.04
EPAN[1]	6.24	5.97	7.13	5.76
DCGN[2]	6.88	6.43	7.68	6.32
Ours	8.05	7.75	8.80	7.70

readability of the sentence, and (4) usability: how useful would the sentence be for a person (especially a low vision person) to understand what is happening in the video. For each of these four metrics, the score ranges from 1 to 10; and the higher the score, the better the performance on the metric. Subsequently, we invite ten participants with excellent English skills to score a subset of 50 video-caption pairs on these four metrics. For the state-of-the-art models and our model, we calculate the average of scores on these four metrics, respectively. As shown in Table XIV, our model is significantly ahead of the state-of-the-art methods in all four metrics, which shows that our model does not simply imitate the reference sentences, but learns the representation of emotional and factual semantics.

F. Qualitative Results

To make a complete comparison with the state-of-the-art methods, we conduct three kinds of qualitative studies.

Firstly, we compare the caption quality with the baseline method DCGN and ground truth. As shown in Fig. 5, it can be seen that our method could generate more vivid descriptions than DCGN. For example, in the last video sample, DCGN completely follows the subjective emotional “excited” but ignores the objective emotional “crying” in the video. In contrast, our model perceives objective emotion increments and provides accurate emotion guidance at each time step.

Secondly, we also compare the performance on different subjective emotional inputs for the same video. As shown in the first video, in Fig. 6, we use “excited” and “scared”

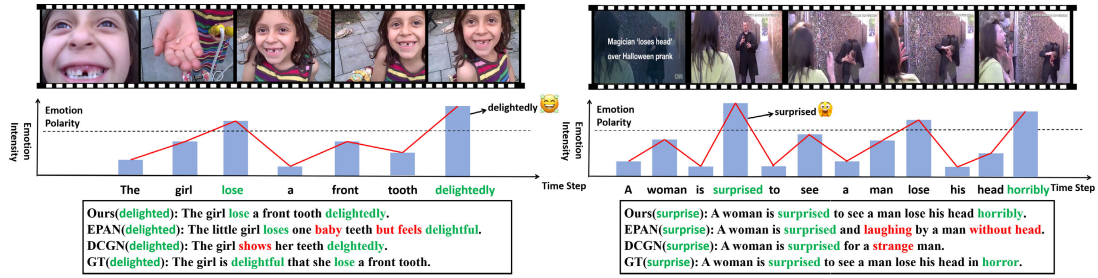


Fig. 5. Qualitative results on EmVidCap-L dataset to compare with the existing method and ground-truth.



Fig. 6. Qualitative results on EmVidCap-L dataset to show the emotional caption generation performance under diverse subjective emotions.

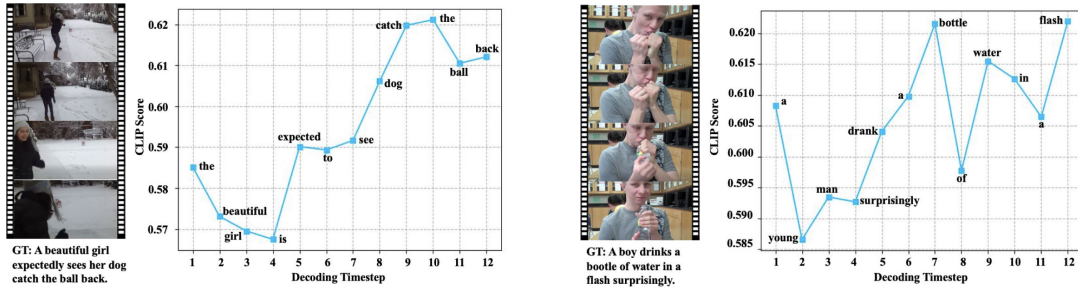


Fig. 7. Visualization of CLIP Scores between partial captions and input video during the decoding timesteps. The CLIP representation of the video is the average of each sample-frame representation.

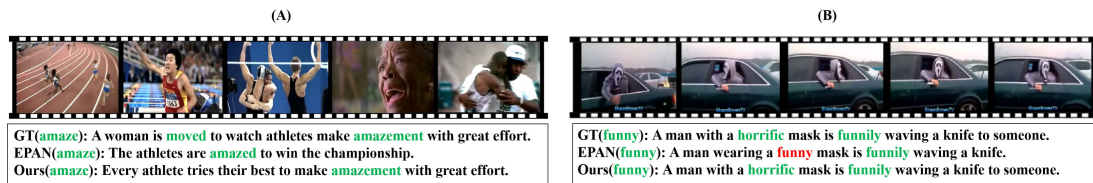


Fig. 8. Visualization of failure cases and hard samples on EmVidCap-H dataset.

as subjective emotions, respectively, to generate two different subjective captions with good quality. Our model makes the emotional style of each generated caption consistent with the input subjective emotion by adapting emotion correlation and disentanglement, which provides diversity for generations.

Furthermore, in order to intuitively display the increment of the semantic richness during the caption generation process, we make a visualization of CLIP Scores between partial sentences and input videos. As shown in Fig. 7, we observe that the CLIP Scores continue to improve overall with time steps. In particular, the CLIP Scores significantly improve when generating concepts (*i.e.*, dog, water) and emotional words (*i.e.*, expected) related to the video content, which shows that the captions generated by our model are not only consistent

with ground truth, but also semantically similar to the video content.

Finally, we add a visualization for hard samples and failure cases to comprehensively analyze our model shown in Figure 8. For Example (A), we add a failure case of our model. our model focuses entirely on the athletes, missing the factual content and emotions of the woman. We analyze that our model lost judgment of the emotional subject due to the rapidly changing scenes and characters. Meanwhile, we believe that the lack of understanding of such complex scenes is a common problem of all current captioning models, and we will continue to work on it in the future. Besides, for Example (B), we add a qualitative result of a hard sample on the EmVidCap-H dataset. The objective emotion is that the man with the ghost mask in the video is horrific. However,

TABLE XV
THE COMPARISON WITH STATE-OF-THE-ART METHODS THAT LEVERAGE ADVANCED VIDEO LVLMS FOR GENERATION

Method	BLEU-1 $\uparrow$	BLEU-4 $\uparrow$	EmVidCap METEOR $\uparrow$	ROUGE-L $\uparrow$	CIDEr $\uparrow$	Acc _{sw} $\uparrow$	EmVidCap-H Acc _c $\uparrow$	BFS $\uparrow$	CFS $\uparrow$
CoCap [82]	75.9	27.7	24.9	54.3	48.7	45.9	44.5	42.3	48.0
RTQ [83]	79.1	28.2	24.5	56.1	49.0	41.8	40.9	42.9	47.5
VideoBLIP [84]	79.4	28.1	24.2	55.8	52.6	54.7	53.0	44.8	52.6
HowToCaption [85]	79.5	28.9	25.5	56.4	51.2	55.7	53.9	46.0	51.9
Ours	79.8	29.5	26.6	58.0	52.9	75.6	73.4	49.4	56.2



Fig. 9. Qualitative results to compare the performance of our method and GPT-4o, a typical Large Vision-Language Model, on SVC task. For example, **(excited)** denotes that **excited** is sent to the models as subjective emotion.

the subjective emotion expressed in the ground-truth sentence is funny, which is in conflict with the objective emotions and is difficult for the model to learn accurate emotional semantics. EPAN [10] struggles to handle this emotionally conflicting situation, resulting in a wrong caption ‘wearing a funny mask’. Therefore, we define such videos as hard samples to evaluate the model’s ability to handle challenging scenarios of emotional conflict.

G. Comparison With Large Vision Language Models

Since existing LVLMS show powerful capabilities in various multimodal tasks, it is necessary to compare our model with LVLMS. We investigate the existing LVLMS that could be used for subjective video captioning and find that they could be divided into two types: 1) existing state-of-the-art methods that leverage advanced LVLMS for generation. They can be fine-tuned on our dataset at a minimal computational cost. 2) Directly call the API of some closed-source large models to generate descriptions. We use appropriate prompts to guide these LVLMS without fine-tuning to generate emotional descriptions.

For the first case, we add a comprehensive comparison with some existing state-of-the-art methods that leverage advanced LVLMS for generation, including CoCap (ICCV 2023) [66], RTQ (ACM MM2023) [83], VideoBLIP [84], and HowToCaption (ECCV2024) [85]. We manually add a subjective emotion input like our model for these methods. As shown in Table XV, they show competitiveness in factual semantic metrics. However, these methods have very poor performance for emotion accuracy metrics on EmVidCap-H dataset. We analyze that on the one hand, these methods lack dedicated modules to capture the objective emotional semantics in videos, and on the other hand, they are difficult to leverage subjective emotional semantics to guide caption generation, which leads to inaccurate generation of emotional words and also reduces the factual semantic metrics to a certain extent. In contrast, our model leverages the Objective Emotion Incremental Learning module to fully explore the implicit objective emotions in the video and effectively integrates subjective and objective

TABLE XVI
THE RESULTS OF COMPARISON BETWEEN OUR MODEL AND SOME LVLMS

Method	BLEU-1 $\uparrow$	BLEU-4 $\uparrow$	METEOR $\uparrow$	ROUGE-L $\uparrow$	CIDEr $\uparrow$
GPT-4o [86]	19.4	0.0	17.8	17.3	11.7
VideoLLaMA3 [87]	22.3	2.6	19.3	20.4	15.9
VideoChat [88]	38.4	5.1	21.1	33.7	28.9
Ours	79.2	29.5	24.3	51.5	45.1

emotions to guide caption generation through the Subjective-Objective Emotion Correlation module.

For the second case, we choose three powerful LVLMS, GPT4-o [86], VideoLLaMA3 [87], and VideoChat [88], and select the subset of EmVidCap-L as the test set. Subsequently, we provide the video samples and give the following prompts:

“I have given you the video and specified the subjective emotion as <Subjective Emotion>. Please generate English captions of no more than 15 words for this video. Please carefully analyze the visual content and factual semantics in the video.”

The results are shown in Table XVI and the visualization results of some samples are shown in Fig. 9, we observe that they perform poorly on the SVC task. We analyze that they are struggle to capture emotional semantics in videos and bridge the gap between the subjective emotional input of the text modality in the prompts and the objective emotions in the visual modality. Besides, they lack an effective aggregation way to collaboratively fuse two emotions, making it difficult to correctly guide caption generation. On the contrary, our method leverages subjective emotional input as the basis to mine objective emotional increments in videos, while using the subjective-objective emotion correlation module to achieve effective emotional aggregation and disentanglement. Finally, the emotion-correlated generation module leverages emotion-adaptive decoder to achieve accurate emotional guidance.

V. CONCLUSION

In this paper, we investigate a new subjective video captioning (SVC) task, which integrates the viewer subjective emotional reactions into video caption generation. To this end,

a novel Subjective-Objective Emotion Correlated Generation Network (SO-ECGN) is proposed for the SVC task, which collaborates subjective and objective emotions to guide adaptive caption generation. To be specific, an objective emotion incremental learning module, including adaptive video reweighting, dynamic mask attention and emotion domain shifting, is designed to mine implicit visual emotional cues based on the subjective emotion. Besides, a subjective-objective emotion correlation module is proposed to provide customized accurate emotion guidance by disentangling into emotion polarity and intensity for each generation step. Extensive experiments on three challenging datasets demonstrate the superiority of our method and each proposed module.

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